LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework 3
Deadline 15/10/2024 23:00
Submit on Fenix as pdf

I) Polynomial Regression (2 pts)

Consider a training set with 5 observations (sample) with dimension $D = 1$

$$x_1=1,\ x_2=0,\ x_3=-2,\ x_4=5,\ x_5=-5$$

With targets

$$t_1=-1,\ t_2=1,\ t_3=-1,\ t_4,=0,\ t_5=-1$$

$$\phi_j(x) = x^j$$

which can lead to a polynomial regression of the third degree

$$y(x, \mathbf{w}) = \sum_{j=0}^{3} w_j \cdot \phi_j(x) = w_0 + w_1 \cdot x + w_2 \cdot x^2 + w_3 \cdot x^3.$$

You can (should?) use your computer with Python, NumPy, MATLAB, Octave, Mathematica, etc. (whatever tool/language you like). Please ***indicate your results step by step****. In the exam the examples will be much simpler, so you can get a solution using a calculator right on time.*

(a) (0.5 pts)

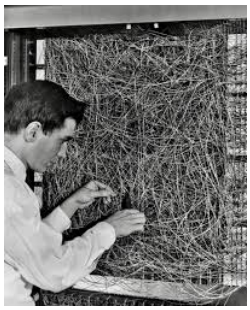
Compute the design matrix Φ.

(b) (1.5 pts)

Compute the polynomial regression weights.

LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework 3
Deadline 15/10/2024 23:00
Submit on Fenix as pdf

II) Perceptron (2pts)



Consider the following linearly separable training set:

$$x_1 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}, x_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, x_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

with the target values

$$t_1=0,\ t_2=0,\ t_3=1$$

Initialize all weights to one (including the bias). Use a learning rate of one for simplicity.

Apply the perceptron learning algorithm with the sgn_0 activation function

$$sgn_0(x) = \begin{cases} 1 \text{ if } & x \geq 0 \\ 0 \text{ if } & x < 0 \end{cases}.$$

for **one** iteration.

III) Neural Network NN (4 pts)

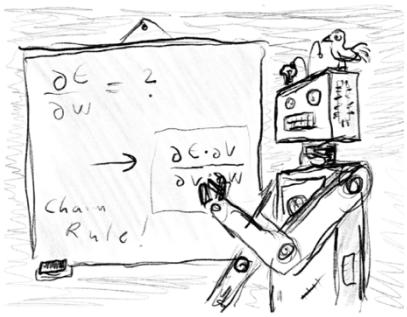
Given the weights

$$W^{[1]} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$b^{[1]} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$W^{[2]} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}$$

$$b^{[2]} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



and the activation function ReLU of the hidden layer and SoftMax of the output layer using the cross-entropy error loss do a stochastic gradient descent update (with learning rate η = 1) for the training example:

$$\mathbf{x}=(1,1,1,1,1)^T \text{ and the target } \mathbf{t}=(1,0)^T,$$

Indicate the new weights and bias.

LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework 3
Deadline 15/10/2024 23:00
Submit on Fenix as pdf

VI) Software Experiments (2pts)

Download the jupyter notebook HM3_25_NN.ipynb.

Split the data using the command (in the notebook)

```
digits = datasets.load_digits()

X, y = digits.data, digits.target

X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.7,
stratify=y, random_state=42)
```

Compare the accuracy on the test set of Logistic Regression with NN.

```
predictor = MLPClassifier(hidden_layer_sizes=(10,4),random_state=42,activation
='logistic',solver='sgd',max_iter=2000)
```

Layer size 10, 4 means two hidden layers, first layer 10 neurons and second hidden layer 4 neurons. Can you improve accuracy on the test set by changing the parameters of hidden_layer_size?

(a) (1.5 pts)

Indicate your best parameters of the hidden layer and output size and compare the results (accuracy) with the logistic regression Indicate the loss curve of both experiments. What is your conclusion (one sentence)?

(b) (0.5 pts)
Using the best architecture change for the activation function 'identity'. Is the result still better? Indicate why not to use the identity function in the hidden layer

Pls, do not spend too much time on the experiments! You should get a feeling what the search for the right parameters for NN models is, not the search for the best model!

NN learning is a **black box** but it also seems "mysterious or magical"

Many approaches and hyperparameters: activation functions, learning rate,